

GERMANS SAVCISENS

Ph.D. · Postdoctoral Research Associate, Northeastern University

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PROFILE

Postdoctoral researcher with a PhD in Applied Mathematics and Computer Science, specializing in large language models, interpretability, and population-scale behavioral modeling. Developed [life2vec](#), a transformer trained on nationwide longitudinal registry data to predict life outcomes across an entire population, and CoevolveSim, a framework for synthesizing and analyzing networks of interacting LLM agents (opinion dynamics). Hands-on Python/PyTorch experience across the full research pipeline, from large-scale data processing to model development, evaluation, and algorithmic auditing. Experienced working in multidisciplinary teams spanning computer science, social science, and statistics, with a track record of mentoring graduate researchers on interpretable machine learning and LLM auditing.

Areas of interest: transformer architectures, LLM interpretability & auditing, population-scale behavioral modeling, agent-based simulation, evaluation of predictive models, epistemology of LLMs, computational social science.

EDUCATION

Technical University of Denmark

2020–2024

PhD, Applied Mathematics and Computer Science

Copenhagen, Denmark

- Dissertation: *Life Trajectories as Symbolic Language: Exploring Human Behaviour with Language Models*
- Committee: Søren Hauberg (chair), Mads Nielsen, Arkadiusz Stopczynski

Technical University of Denmark

2018–2020

MSc, Human-Centered Artificial Intelligence

Copenhagen, Denmark

Aalto University

2021

Visiting Student, Digital Ethics & Statistical Natural Language Processing

Helsinki, Finland

Aalborg University

2013–2016

BSc, Human-Computer Interaction

Copenhagen, Denmark

RESEARCH EXPERIENCE

Northeastern University

May 2024–present

POSTDOCTORAL RESEARCH ASSOCIATE | Mentor: Tina Eliassi-Rad

Boston, USA

- Developed LLM probing frameworks for knowledge representation and belief stability; open-source auditing codebase [\[sAwMIL\]](#).
- Designed and implemented a multi-agent framework in PyTorch to study opinion dynamics in networks of LLM-based agents [\[CoevolveSim\]](#).
- Supervised graduate researchers in interpretable machine learning, and LLM auditing, and coordinated research on multi-agent interactions.
- Presented projects at the NeurIPS Mechanistic Interpretability Workshop, IC2S2, and NEMI Workshop; submitted three manuscripts for peer review at leading venues.

Technical University of Denmark & University of Copenhagen

Sep 2020–Apr 2024

PHD STUDENT & RESEARCH ASSISTANT | Advisors: Sune Lehmann, Lars Kai Hansen

Copenhagen, Denmark

- Led population-scale modeling of nationwide socioeconomic and health trajectories using transformer-based models trained on longitudinal registry data [\[life2vec\]](#).
- Developed and evaluated life-outcome prediction models, with emphasis on interpretability and uncertainty analysis.
- Taught and developed course materials for a graduate-level course in Social Network Science.
- Invited talks at MIT, University of Oxford, and NordicAI.

Statistics Denmark, Data Science Lab

Sep 2020–Apr 2024

GUEST RESEARCHER

Copenhagen, Denmark

- Collaborated with economists and social scientists to conduct quantitative and qualitative analyses of demographic and socioeconomic patterns.
- Led modeling of large-scale longitudinal datasets.

Network Science Institute, Northeastern University

Sep 2022–Feb 2023

VISITING PHD STUDENT

Boston, USA

- Analyzed ethical and societal risks of deep learning applied to socioeconomic data.
- Investigated methods for enhancing transparency, accountability, and fairness in algorithmic systems

SELECTED PUBLICATIONS

Savcicens, G., Dies, S., Maynard, C., Eliassi-Rad, T. *Belief Coevolution in a Social Network of Generalist and Specialist Large Language Models*. Technical Report, 2026. [\[Report\]](#) [\[Code\]](#) [\[NetSci Poster\]](#)

Savcicens, G. & Eliassi-Rad, T. *Trilemma of Truth in Large Language Models*. Mechanistic Interpretability Workshop, NeurIPS, 2025. [\[Latest\]](#) [\[OpenReview\]](#) [\[Code\]](#) [\[NeurIPS Poster\]](#)

Savcicens, G. *Large Language Models Act As If They Are Part of a Group*. Nature Computational Science, 2025. [\[DOI\]](#)

Dies, S., Maynard, C., **Savcicens, G.** & Eliassi-Rad, T. *Representational and Behavioral Stability of Truth in Large Language Models*. arXiv, 2025. [\[Latest\]](#)

Savcicens, G., Eliassi-Rad, T., Hansen, L.K., Mortensen, L.H., Lilleholt, L., Rogers, A., Zettler, I. & Lehmann, S. *Using Sequences of Life-Events to Predict Human Lives*. Nature Computational Science, 2024. [\[DOI\]](#) [\[Official Page\]](#)

Fernández, E. F. & **Savcicens, G.** *A Sustainable West? Analyzing Clusters of Public Opinion in Sustainability*. Digital Humanities in the Nordic and Baltic Countries, 2023. [\[DOI\]](#)

TEACHING & MENTORING

Northeastern University 2024–2026

GUEST LECTURER | *Foundations of AI (2026); Machine Learning with Graphs (2024 & 2025)*

IT University of Copenhagen 2022–2023

PRINCIPAL LECTURER (2023); ASSISTANT LECTURER (2022) | *Algorithmic Fairness, Accountability and Ethics*

Technical University of Denmark 2019–2021

TEACHING ASSISTANT | *Advanced Machine Learning; Social Data Analysis; Social Graphs and Interactions*

Mentoring & Supervision

Co-supervised 5 MSc thesis and term projects on NLP for legal/compliance documents and longitudinal app-usage modeling, Technical University of Denmark (2021–2024).

SELECTED TALKS & PRESENTATIONS

Belief Coevolution in a Social Network of LLMs Jun 2026
International Conference on Network Science (Poster + Talk) Boston, USA

Formal Epistemology of Large Language Models Apr 2026
Social Complexity Lab, University of Copenhagen (Talk) Copenhagen, Denmark

From Human Lives to Machine Minds [\[Slides\]](#) Apr 2026
Rising Stars in Computational & Data Science, Santa Fe Institute (Workshop + Talk) Santa Fe, USA

Life2vec: Foundation Models for Registry Data [\[Slides\]](#) Nov 2025
Networks for Healthy Aging Workshop (Invited Talk) Vienna, Austria

From Life-Courses to Representations: Foundation Models for Registry Data Sep 2025
SWECOV Workshop (Keynote Speaker) Stockholm, Sweden

LLMs for Health and Labor Data Jan 2025
Max Planck Institute (Invited Speaker) Rostock, Germany

Life2vec: Life Trajectories in High-dimensional Spaces [\[Recording\]](#) Nov 2024
Chinese University of Hong Kong (Invited Speaker) Hong Kong, HK (Remote)

Learning Representations of Socio-economic Data Jan 2024
University of Oxford (Invited Speaker) Oxford, UK (Remote)

PROFESSIONAL SERVICE

SESSION CHAIR: International Conference on Network Science (NetSci), ML & AI track, 2026

JOURNALS: Nature Computational Science, Artificial Intelligence, Science Advances, Scientific Reports, npj Complexity, EPJ Data Science

CONFERENCES & WORKSHOPS: NeurIPS (Ethics Reviewer 2023–2026; Mechanistic Interpretability Workshop 2025), IC2S2 2025, EMNLP 2024, Conference on Complex Networks and Their Applications 2025

AWARDS & GRANTS

Rising Stars in Computational and Data Sciences 2026 2026
Santa Fe Institute | Nominated Participant USA

Travel Grants 2022
Otto Mønsted Fond, STIBO Foundation, William Demant Fonden, Denmark Denmark